

Sustainable Synthesis of Acyl Fluoride via 1,3-Chelation-Assisted Transhalogenation and Its Application in the Valorization of Biomass-Derived Alcohol in Aqueous System

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Abstract: We developed an environmentally friendly method for synthesizing acyl fluorides and achieving the selective mono-acylation of water-soluble alcohols using acyl fluorides in green media. Our method offers substantial advantages over the existing technologies for acyl fluoride production, particularly in terms of cost and environmental impact. Notably, our approach is augmented by an unusual 1,3-chelation via transhalogenation, and the use of an eco-friendly solvent and a simple base aligns with the principles of sustainable chemistry. We also explored the direct and highly efficient mono-acylation of biomass-derived water-soluble alcohols, such as ethylene glycol and glycerol, in the presence of water. The key arrangements, such as the role of the K_2CO_3 base and unusual 1,3-chelation between acyl fluoride and potassium ion, were characterized by HRMS and NMR spectroscopic analysis, and controlled experiments. These findings underscore the effectiveness of this method and provide deeper insights into the reaction mechanism.

Keywords: fluorination; acyl fluoride; transhalogenation; 1,3-chelation; biomass valorization

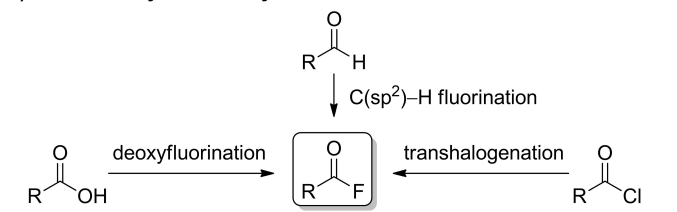
Introduction

Since organofluorine moieties are crucial in pharmaceuticals,^[1] pesticides,^[2] and other materials,^[3] an increasing attention has been directed towards developing synthetic methods that facilitate the incorporation of fluorine or fluorinated groups into organic molecules.^[4] Among organofluorine compounds, acyl fluorides have recently gained significant attention as building blocks in transition-metal catalysis owing to their potential to serve as at least three distinct synthetic fragments in organic chemistry;^[5] namely ‘RCO’ source, ‘R’ source, and ‘F’ source.^[6] In addition, acyl fluorides offer distinct advantages in synthetic applications over acyl chlorides, which are valuable synthons known for their unique balance between stability and reactivity, specifically their stability in water.^[7] Consequently, we focused on developing new synthetic routes for acyl fluoride production.

A general route for the synthesis of acyl fluorides involves nucleophilic addition to activated carboxylic

acids (deoxyfluorination),^[8] transhalogenation (or halogen exchange reaction) with acyl chlorides,^[9] and $C(sp^2)$ -H fluorination (Scheme 1a).^[10] Among these synthetic methods, the conversion of carboxylic acids to acyl fluorides represents the most straightforward approach, involving deoxyfluorinating agents such as cyanuric fluoride, fluoro formamidinium salts (TFFH), DAST, Deoxo-Fluor, $[R_2N=SF_2]BF_4$ (XtalFluor), $[Me_4N]SCF_3$, and $PPh_3/NBS/Et_3N \cdot HF$. In the context of transhalogenation, two distinct methodologies are employed: the use of transition-metal complexes and a conventional approach involving the use of a metal fluoride as an F^- source. The method involving transition-metal complexes can be further subdivided into stoichiometric approaches, such as $[Cp_2Co]F$ and $Cp^*Ir(Ph)(F)(PMe_3)$,^[9i-k] and catalytic methods such as $CpCo(I)_2(PPh_2Me)$ in conjunction with AgF .^[9] Conventional methods involve using inorganic salts such as KHF_2 , spray-dried KF , $KF/TEBA$, NaF , or CaF_2-KF , which present several limitations, including the need for extensive drying, prolonged reaction times

a) Common acyl fluoride syntheses

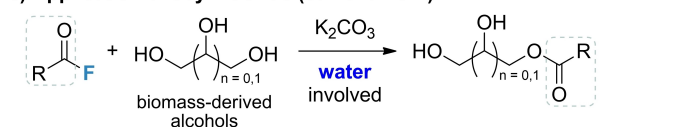


b) This work



- ✓ Employing the simplest F^- source
- ✓ Using eco-friendly solvent
- ✓ **Synergistic acceleration of reaction rate** through unusual 1,3-chelation and ion-dipole interaction, as confirmed by NMR spectroscopy
- ✓ **Transition-metal-free** variant and complementary approach to the use of either cobaltocenium fluoride or perfluoroalkyl cobalt(III) fluoride, both of which serve as effective for generating naked fluoride and facilitating transhalogenation^[Ref. 9+9]

c) Application of acyl fluoride (current work)



- ✓ **Transition-metal-free** version
- ✓ The perfect arrangement for accelerating the reaction, involving simple base and unusual 1,3-chelation between acyl fluoride and potassium ion, in the presence of water
- ✓ Elucidating the key intermediate using HRMS and NMR spectroscopic analysis

Scheme 1. Synthesis of acyl fluorides includes: (a) previous approaches, (b) this study employing an unusual 1,3-chelation and ion-dipole interaction strategies, and (c) their application to the selective mono-acylation of biomass-derived alcohols in the presence of water.

at elevated temperatures, and the requirement for fractional distillation to isolate products.^[11]

For $C(sp^2)$ -H fluorination, aldehydes (or alcohols) in conjunction with reagent systems such as $FOSO_2OCs$ or *N*-fluorobenzenesulfonimide (NFSI) under light irradiation are employed. Therefore, there is still room for improvement in the synthesis of acyl fluorides, particularly in terms of eco-friendliness, efficiency, practicality, and simplicity.

Herein, we report serendipitous discoveries on the synergistic acceleration of the reaction rate for acyl fluoride synthesis, utilizing the simplest F^- source and an environmentally benign acetone solvent. Specifically, this phenomenon is significantly enhanced by the unprecedented 1,3-chelation^[12] between acyl fluoride and K^+ ion, coupled with ion-dipole interaction between K^+ ion and oxygen atom in acetone. This approach facilitates transhalogenation, even when employing unreactive alkali metal fluoride without the assistance of activators (such as transition-metal complexes or catalysts). Thus, this study presents a complementary strategy for the stoichiometric or

catalytic use of either cobaltocenium fluoride or perfluoroalkyl cobalt(III) fluoride. Both methods efficiently generate naked fluoride ions and serve as potent sources for transhalogenation. Additionally, the complete dissociation of KF, facilitated by these two key interactions, contributes to the acceleration of the reaction (Scheme 1b). Furthermore, we investigated the application of the synthesized acyl fluorides in conjunction with water-compatible biomass-derived alcohols for valorization (specifically, selective mono-acylation) in the presence of water (Scheme 1c). To attain reaction selectivity, existing methods predominantly employ transition-metals (e.g., Sn and Fe),^[13] either stoichiometrically or catalytically, to disrupt inter-/intramolecular interactions (specifically hydrogen bonding) through chelation with diols, thereby increasing the solubility in organic solvents and facilitating mono-acylation reactions.^[14] In contrast to existing techniques, our newly designed protocol offers several advantages, as follows: i) elimination of the use of precious transition-metals; ii) perfect arrangement, marked by the use of a simple base such as K_2CO_3 and an unusual 1,3-chelation between acyl fluoride and K^+ ion, which significantly accelerates the reaction rate in aqueous system by eliminating the inter-/intramolecular H-bonds of diols or polyols; iii) the key step and intermediate are elucidated through NMR spectroscopy and HRMS analysis.

Results and Discussion

Our investigation commenced with the use of benzoyl chloride **1a** as a model substrate to optimize the transhalogenation conditions. Metal fluorides were employed as the F^- source in various solvents at room temperature (Table 1). KF (2 equiv.) was dissolved in halogenated solvents such as CH_2Cl_2 and $CHCl_3$ and stirring at room temperature for 9 h to obtain benzoyl fluoride **2a** as the desired product in 35% and 39% yields, respectively (Table 1, entries 1–2). The use of THF resulted in a product yield comparable to that obtained using halogenated solvents (Table 1, entry 3). Replacing THF with 1,4-dioxane decreased the product yield, whereas the use of non-polar toluene as the solvent increased the product yield to 58% (Table 1, entries 4–5). A substantial enhancement in product yield (90%) was achieved using acetone as an eco-friendly solvent (Table 1, entry 6). The reaction performed using 3 equiv. of KF did not improve the yield (Table 1, entry 7). Alternatives to KF such as NaF and CsF showed effectiveness, but with diminished yields. NaF was ineffective because of its insolubility in acetone (Table 1, entry 8), whereas CsF was effective, albeit with a reduced yield (Table 1, entry 9). Furthermore, increasing the reaction temperature from room temperature to 40 °C resulted in a significant decrease in yield, with the product yield dropping to 21%

Table 1. Optimization of reaction conditions.^[a]

$\text{Ph-C(=O)Cl} \xrightarrow[\text{solvent, RT}]{\text{F}^- \text{ source}} \text{Ph-C(=O)F} \text{ via transhalogenation}$				
Entry	F ⁻ source [equiv.]	Solvent	Time [h]	Yield [%] ^[b]
1	KF [2]	Dichloromethane	9	35
2	KF [2]	Chloroform	9	39
3	KF [2]	Tetrahydrofuran	9	38
4	KF [2]	1,4-Dioxane	9	17
5	KF [2]	Toluene	9	58
6	KF [2]	Acetone	9	90
7	KF [3]	Acetone	9	60
8	NaF [2]	Acetone	9	n.d.
9	CsF [2]	Acetone	9	35
10 ^[c]	KF [2]	Acetone	9	21

^[a] General conditions: benzoyl chloride **1a** (0.5 mmol), F⁻ source (2 equiv.), solvent (0.2 M), RT.

^[b] Yields of the products were determined by ¹⁹F NMR spectroscopy using 2-chloro-4-fluorotoluene as an internal standard.

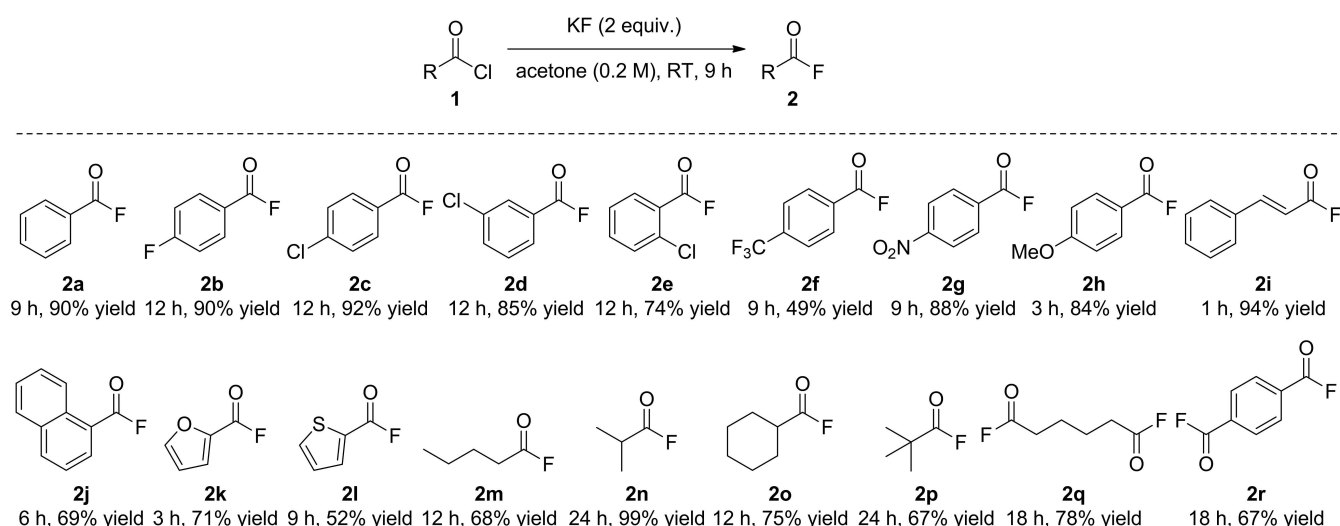
^[c] Reaction at 40 °C.

(Table 1, entry 10). The lower yield (21%) of the desired product **2a** can be attributed to the presence of unreacted benzoyl chloride in the reaction mixture at 40 °C. No by-product was observed, indicating that this temperature is critical for this transformation.

After establishing the optimal reaction conditions, we investigated the influence of acyl chloride derivatives (**1a–1r**) with different substituents under the optimized conditions. The results are shown in Scheme 2. Various aryl-functionalized acyl chlorides

were effectively utilized for the synthesis of acyl fluorides, yielding good to excellent results (74–94%), except for *p*-trifluoromethyl-substituted aryl groups on the acyl chlorides (**1f**). The yield (94%) of *trans*-cinnamoyl chloride **1i** was superior when compared to that of benzoyl chloride **1a**. Heteroatom-containing 2-furanoyl and 2-thenoyl chlorides yielded the desired products **2k** and **2l**, respectively, with efficiencies of 71% and 52%. Various acyl chlorides with distinct aliphatic groups, including *n*-butyl (**1m**), iso-propyl (**1n**), cyclohexyl (**1o**), and *t*-butyl (**1p**) were also evaluated. Although these compounds were more sensitive to moisture than their aromatic counterparts, the corresponding acyl fluoride derivatives were successfully formed (**2m**, **2n**, **2o**, and **2p**, respectively) in good yields. The methods for preparing adipoyl fluoride (**2q**) and terephthaloyl fluoride (**2r**), which serve as invaluable intermediates in the synthesis of nanofiltration membranes, polyamide materials, biodegradable poly(anhydride-ester) polymers, conducting polymers, as well as pesticides and pharmaceuticals, are delineated. Despite their broad range of applications, only a limited number of synthetic methodologies have been documented, and these processes typically require harsh conditions. However, under our optimal conditions, adipoyl fluoride (**2q**) and terephthaloyl fluoride (**2r**) were obtained in good yields, surpassing those of the existing reactions. We then redirected our research focus to the utilization of acyl fluorides in selective organic transformations, leveraging the robust concept of acyl fluoride stability in aqueous systems.

Next, we investigated the feasible reaction conditions for biomass valorization (specifically monoacylation) of water-compatible biomass-derived alco-



Scheme 2. Reaction scopes. The yields of products **2** were determined by ¹⁹F NMR spectroscopy using 2-chloro-4-fluorotoluene as an internal standard.

hol, such as glycerol, in aqueous systems, in the combination with appropriate acyl fluorides (Table 2).

In the initial experiments, we performed the reaction using pyridine as both the solvent and base, which resulted in a high selectivity (**4a/5a** = 54:42) and moderate yield (Table 2, entry 1). Although pyridine offers high selectivity for the mono-acylated product **4a**, its use presents significant environmental hazards and poses an obstacle to extensive industrial applications. To replace the harmful pyridine, we initially screened environmentally beneficial bases. We tested K₂CO₃ as a base, considering the environmental concerns across various solvents. Among the different solvents, THF exhibited excellent selectivity for the mono-acylated product, specifically favoring the primary alcohol over the secondary alcohol in glycerol, and achieved a good yield (Table 1, entry 2). Acetone and water also showed excellent selectivity for the desired products, albeit with reduced yields (Table 2, entries 4–5). To pursue sustainable and green chemistry, we opted for a co-solvent system comprising acetone and water, owing to its miscibility and excellent selectivity for the mono-acylated product. After a thorough evaluation of several parameters, including the acetone-to-water volume ratio, type of base, and base equivalent, the highest chemical yield of 98% was achieved while maintaining excellent product selectivity (Table 2, entry 9). To demonstrate

the robust stability of acyl fluoride in aqueous system, we conducted a reaction using benzoyl chloride **1a** instead of benzoyl fluoride **2a** under optimal conditions. This resulted in a significantly reduced yield (42%) and the formation of undesired by-product such as benzoic acid **7a**, which arose from the hydrolysis of benzoyl chloride (Table 2, entry 10). These findings indicate that acyl fluoride is a superior acylating agent for aqueous systems. By changing the concentration and base, a significantly lower product yield was obtained (Table 2, entries 11–15).

We then investigated the reaction scope under the optimized conditions (Table 2, entry 9). As depicted in Scheme 3, electron-withdrawing groups (such as –F, –Cl, –CF₃, and –NO₂) at *ortho*, *meta*, and *para* positions on the acyl chloride substrates were well tolerated, yielding the desired products (**4b–4g**) as the predominant components in yields exceeding 99%. However, *p*-fluoro-, *p*-trifluoromethyl-, and *m*-chloro-benzoyl fluorides exhibit selectivity for di-acylated products (**5b**, **5d**, **5f**) over mono-acylated products (**4b**, **4d**, **4f**), which is presumably attributed to the electronic nature of electron-withdrawing substituents on the aromatic rings of the acyl fluorides. By contrast, a substrate containing electron-donating group (such as –OCH₃) exhibited diminished reactivity, resulting in lower product (**4h**). Acyl fluorides containing thiophene and furan rings were subjected to biomass

Table 2. Optimization of reaction conditions.^[a]

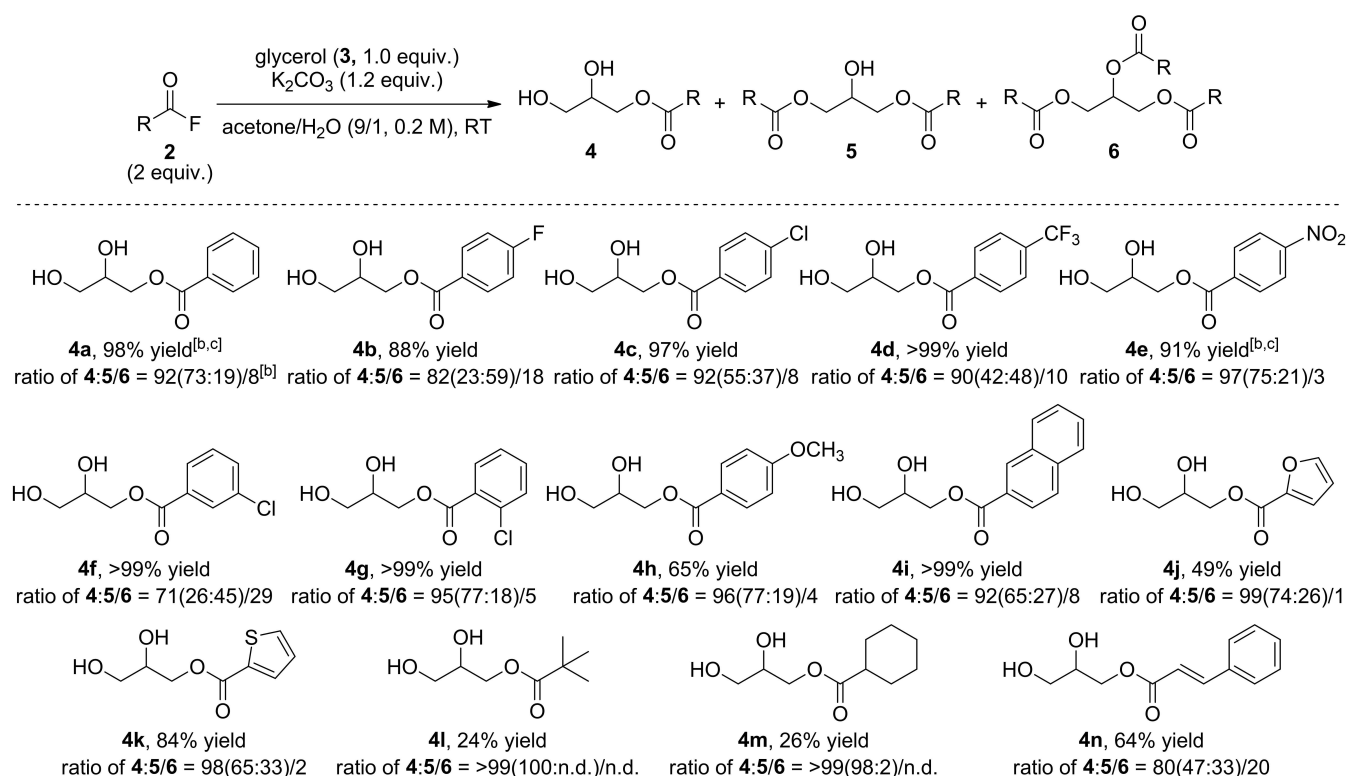
Entry	Solvent	Base [equiv.]	Acylation ratio of 1° alcohol (4a/5a):2° alcohol (6a): 7a ^[b]	Yield [%] ^[b,c]
1	0.2 M Pyridine		96 (54/42):4:0	77
2	Tetrahydrofuran	K ₂ CO ₃ [1.0]	98 (80/18):2:0	82
3	Toluene	K ₂ CO ₃ [1.0]	94 (72/22):6:0	33
4	Acetone	K ₂ CO ₃ [1.0]	99 (86/12):1:0	38
5	Water	K ₂ CO ₃ [1.0]	95 (85/13):5:0	14
6	Acetone/water (v/v = 1:9)	K ₂ CO ₃ [1.0]	98 (85/13):2:0	17
7	Acetone/water (v/v = 1:1)	K ₂ CO ₃ [1.0]	95 (80/15):5:0	23
8	Acetone/water (v/v = 9:1)	K ₂ CO ₃ [1.0]	96 (76/20):4:0	49
9	Acetone/water (v/v = 9:1)	K₂CO₃ [1.2]	92 (73/19):8:0	98
10 ^[d]	Acetone/water (v/v = 9:1)	K ₂ CO ₃ [1.2]	51 (46/6):n.d.:49	42
11	Acetone/water (v/v = 9:1)	K ₂ CO ₃ [2.0]	76 (45/30):24:0	81
12 ^[d]	Acetone/water (v/v = 9:1)	K ₂ CO ₃ [2.0]	77 (67/9):n.d.:23	42
13	Acetone/water (v/v = 9:1)	Li ₂ CO ₃ [1.2]	99 (79/20):1:0	79
14	Acetone/water (v/v = 9:1)	NaOH [1.2]	61 (38/23):39:0	69
15	Acetone/water (v/v = 9:1)	Pyridine [1.2]	> 99 (85/15):0:0	38

^[a] General conditions: glycerol **3** (1 mmol), benzoyl fluoride **2a** (2 mmol, 2 equiv.), solvent (0.2 M), RT.

^[b] The yields and ratio of the products were determined by ¹H NMR spectroscopy using 1,2,4,5-tetramethylbenzene (durene) as an internal standard.

^[c] All yields were for the acylated products.

^[d] Benzoyl chloride **1a** was used instead of benzoyl fluoride **2a**.



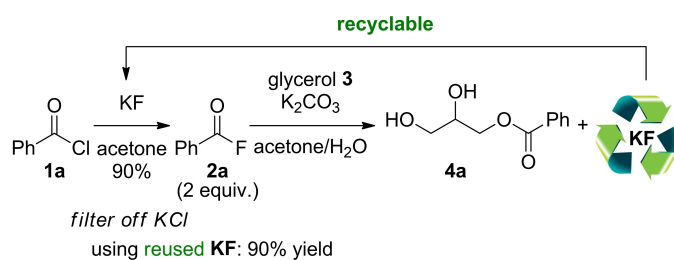
Scheme 3. Reaction scopes.^[a] [a] The acylation reactions were carried out under the following optimal conditions: glycerol **3** (1 mmol), acyl fluoride **2** (2 mmol, 2 equiv.), K₂CO₃ (1.2 mmol, 1.2 equiv.), solvent (0.2 M), RT, 24 h. [b] Yields and ratio of the products were determined by ¹H NMR spectroscopy using 1,2,4,5-tetramethylbenzene (durene) as an internal standard. [c] All yields were for the acylated products. All acyl fluorides were purified prior to use.

valorization using glycerol, with the former (**4k**, 84% yield) ultimately exhibiting superior performance in terms of chemical yield compared to the latter. Aliphatic acyl chlorides were also applicable in this study, leading to the formation of the corresponding products (**4l**, **4m**) in satisfactory yields, which have not been previously reported in the literature. Notably, cinnamoyl chloride afforded the desired product (**4n**) in a higher yield than other aliphatic acyl chlorides. Overall, the reaction exhibits excellent yield and an remarkably accelerated rate, which can be attributed to: i) the perfect arrangement involving the use of a simple base such as K₂CO₃ and the unusual 1,3-chelation between acyl fluoride and K⁺ ions and ii) the ‘in the presence of water effect’,^[15] analogous to the ‘on-water effect’,^[16] which facilitates the creation of a biphasic system. This system was initially formed by the addition of glycerol with K₂CO₃, followed by the formation of an emulsion upon the addition of benzoyl fluoride to acetone/water (v/v = 9:1) mixture (see Supporting Information).

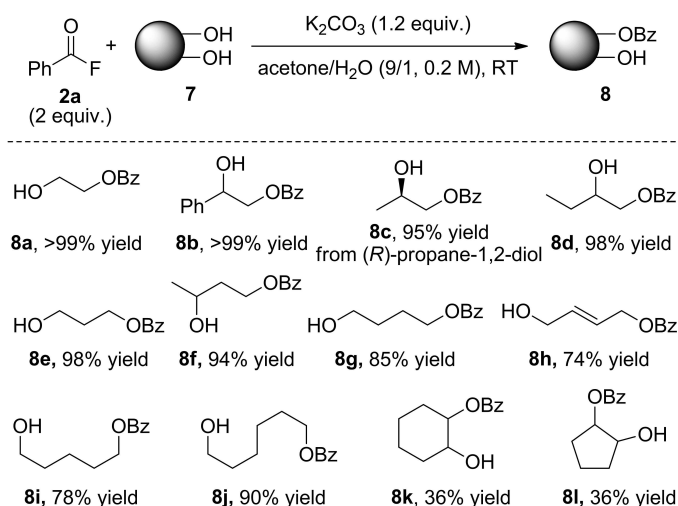
In addition, we postulated the feasibility of recycling KF after the completion of step-by-step reactions, which consist of transhalogenation and acylation processes,^[17] starting from acyl chloride **1a** in conjunction with water-compatible biomass-derived glycerol **3** to synthesize the mono-acylated product **4a** as the predominant component. Indeed, the newly designed approach successfully produced results equivalent to those achieved under the original conditions, thereby enhancing the sustainability and robustness of the chemical process (Scheme 4).

We further investigated a range of water-compatible diols, including biomass-derived ethylene glycol derivatives, with acyl fluoride as the counterpart, and the results are presented in Scheme 5. Notably, nearly quantitative yields and selectively mono-acylated products (**8a–8d**) were achieved using ethylene glycol and its derivatives, including the chiral 1,2-diol. Furthermore, the elongation of the carbon chain in the

erol **3** to synthesize the mono-acylated product **4a** as the predominant component. Indeed, the newly designed approach successfully produced results equivalent to those achieved under the original conditions, thereby enhancing the sustainability and robustness of the chemical process (Scheme 4).



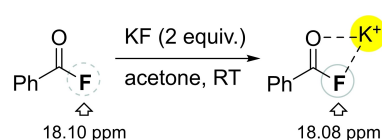
Scheme 4. Recycling of KF.



Scheme 5. Reaction scopes (benzoyl fluoride **2a** was purified prior to use).

ethylene glycol derivatives—such as with propanediol, butanediol, pentanediol, and hexanediol—consistently resulted in high product yields (**8e–8j**). However, cyclic diols afforded the desired products (**8k**, **8l**) in relatively low yields, presumably because of their conformational rigidity.

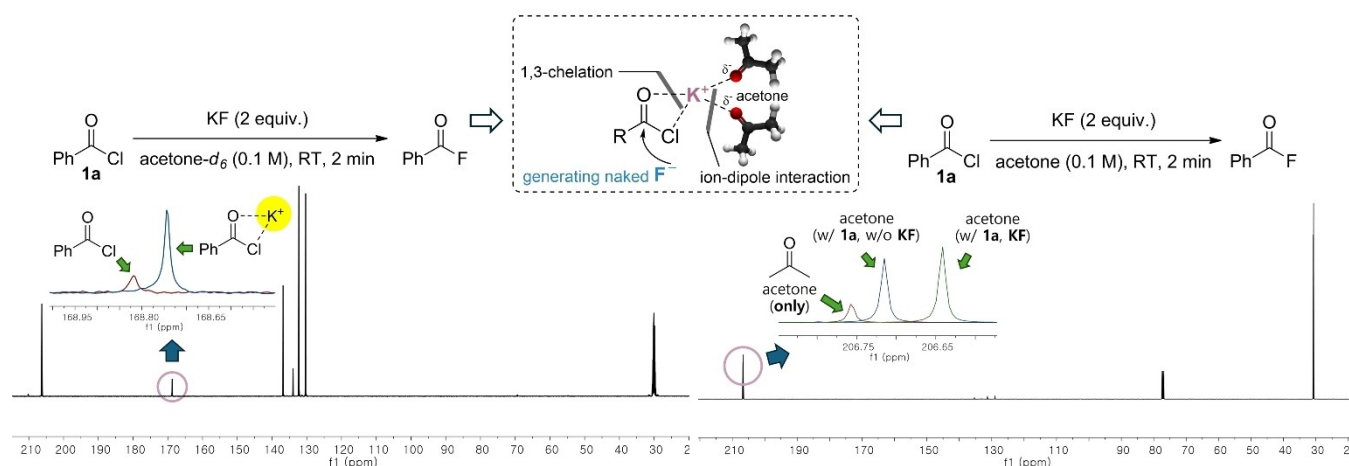
To elucidate the driving force underlying the acceleration of the transhalogenation or selectivity of the acylation reaction, we performed a detailed investigation using NMR spectroscopy and high-resolution mass spectrometry (HRMS). For the transhalogenation reaction, benzoyl chloride **1a** was treated with an equimolar amount of KF in acetone-*d*₆ (1.0 M) at room temperature for 2 min, resulting in an up-field chemical shift of the carbonyl carbon peak in **1a**. This



Scheme 7. Identification of 1,3-chelation between potassium ion and acyl fluoride (instead of acyl chloride) through the measurement of the fluorine atom chemical shifts via ¹⁹F NMR spectroscopy using 2-chloro-4-fluorotoluene as the internal calibrant (100.76 ppm).

observed shift provides compelling evidence that the carbonyl carbon of **1a** is activated by the Lewis acidic K⁺ ion through the unusual 1,3-chelation between the K⁺ ion and acyl chloride moiety. In addition, acetone exhibited an upfield chemical shift in the carbonyl carbon peak of acetone in the presence of **1a** and KF. Based on the solid evidence of two distinct upfield chemical shift phenomena, we propose that the carbonyl carbon of **1a** is highly activated through the unusual 1,3-chelation and ion-dipole interactions (Scheme 6). To further verify the 1,3-chelate formation between acyl fluoride and potassium ions, we conducted a reaction of benzoyl fluoride with potassium fluoride in acetone. The upfield chemical shift of fluorine (18.08 ppm) relative to benzoyl fluoride (18.10 ppm), referenced against the peak of 2-chloro-4-fluorotoluene (100.76 ppm), observed through ¹⁹F NMR spectroscopy, indicating that the fluorine in benzoyl fluoride is also activated by the potassium ion (Scheme 7 and see Supporting Information).

This activation facilitates the release of a naked F[−] ion, thereby enhancing the transhalogenation rate in synergy with the three-component system (acyl chloride, KF, and acetone).

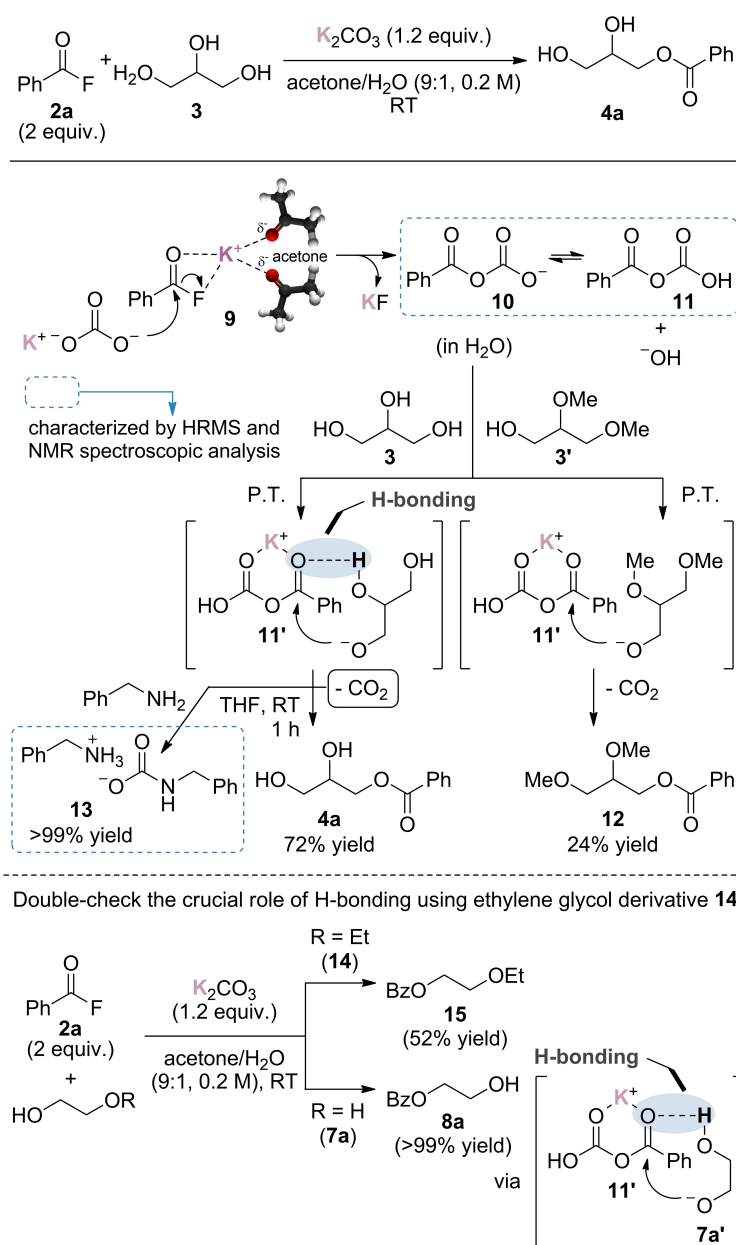


Scheme 6. Identification of unusual 1,3-chelation between potassium ion and acyl chloride, and ion-dipole interactions between potassium-chelated acyl chloride and acetone by ¹³C NMR spectroscopy. Detailed measurement conditions for 1,3-chelation and ion-dipole interactions are provided in the Supporting Information.

To elucidate the reaction mechanism of selective mono-acylation, we reacted acyl fluoride **2** and glycerol **3** in a (CD₃)₂CO/D₂O (9:1, 1.0 M) mixture at room temperature for 2 h. Through high-resolution mass spectrometry (HRMS) analysis and NMR spectroscopy, key intermediates were identified, offering insights into the possible reaction pathways (Scheme 8). Initially, benzoic carbonic anhydride **11** and its anionic form **10**, which existed in equilibrium, were rapidly formed within 1 h by the reaction of highly activated acyl fluoride **9** with potassium carbonate. Two different transient intermediates, **10**

and **11**, were identified via the HRMS (ESI, *m/z*) analysis. For **10**, calcd. for C₈H₅O₄Na [M+Na]⁺ 188.0005; found: 188.0008; for **11**, calcd. for C₈H₆O₄Na [M+Na]⁺ 189.0163; found: 189.0168.

The next step involved a nucleophilic acyl substitution between intermediate **11** and a negatively charged oxygen nucleophile derived from glycerol. To verify the actual activator in this step, we modified the backbone of glycerol **3**, and synthesized its derivative **3'**, and subsequently performed the mono-acylation reaction. However, the desired product **12** was obtained only in low yield (24%) when glycerol 1,2-



Scheme 8. Possible reaction mechanism involving the identification of intermediates through HRMS and NMR spectroscopic analysis. P.T. = proton transfer.

dimethyl ether was employed, whereas a high yield (72%) of the desired product **4a** was achieved using glycerol **3**. The 24% yield obtained from **3'** was attributed to a single, specific activation mode, likely involving chelation (**11'**) between K^+ ion and two carbonyl groups in benzoic carbonic anhydride **11**,^[18] which facilitated the progression of this transformation. Conversely, the higher yield (72%) of the desired product **4a** compared to that of product **12** can be attributed to chelation (**11'**) and, presumably, an additional secondary interaction between the carbonyl oxygen atom in the chelated benzoic carbonic anhydride (**11'**) and hydrogen atom of the glycerol anion via H-bonding. As illustrated in Scheme 8, the CO_2 emission process was investigated by collecting the CO_2 gas released during decarboxylation at the end of the reaction using a balloon. Subsequently, the reaction was performed with benzylamine, leading to the formation of phenylmethanaminium benzylcarbamate **13** in quantitative yield (>99%). The resulting product **13** was characterized by 1H and ^{13}C NMR spectroscopy, and its physical and spectroscopic properties were consistent with those reported in literature for this compound (for details, see Supporting Information).^[19]

To further confirm the crucial role of H-bonding as a secondary interaction, as illustrated in Scheme 8 (above), an ethylene glycol derivative such as 2-ethoxyethanol (**14**) was subjected to benzoylation with benzoyl fluoride **2a** under optimal conditions. As a result, the corresponding product **15** was obtained in 52% yield. The significantly lower yield (52%) from 2-ethoxyethanol (**14**), compared to the >99% yield obtained with ethylene glycol as shown in Schemes 5 and 8 (see product **8a**), implies that hydrogen bonding plays a pivotal role in enhancing the yield (Scheme 8, below).

Overall, benzoic carbonic anhydride **11**, which resulted from the perfect arrangement between acyl fluoride and carbonate base, played a pivotal role as a pre-activator, thereby exerting a substantial influence on the entire reaction for the following reasons: i) retarding the rate of hydrolytic decomposition of acyl fluoride to prevent the undesired formation of carboxylic acid via the hydrolysis of acyl fluoride and ii) significantly enhancing the product yield via complexation with K^+ ions, which facilitates nucleophilic acyl substitution with greater efficiency.

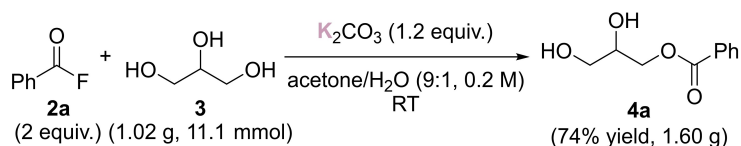
To demonstrate the sustainability of the current protocols, we further expanded for the gram-scale synthesis **4a** (Scheme 9). The reaction of glycerol **3** was scaled up to 1.02 g (11.1 mmol), representing an 11-fold increase over the scale outlined in Scheme 3. It was conducted with acyl fluoride **2a** (2 equiv.) in a mixture of acetone and H_2O (9:1, 0.2 M) at room temperature for 24 h under optimal basic conditions [K_2CO_3 (1.2 equiv.)]. A comparable yield of the desired product **4a** was obtained, as described in Scheme 3, thereby promoting sustainable chemistry, minimizing environmental impact, and maintaining optimal performance.

Conclusion

In summary, we demonstrated an atom-economical and environmentally friendly synthetic method for acyl fluorides and distinguished it from previously established approaches. Our approach to acyl fluoride synthesis is based on an unusual 1,3-chelation via the transhalogenation of acyl chloride, in line with the principles of sustainable chemistry. We also extended the applicability of the synthesized acyl fluorides to the valorization of water-compatible biomass-derived alcohols, including ethylene glycol and glycerol, in the presence of water under perfect arrangement conditions involving a simple base and the unique 1,3-chelation between acyl fluoride and potassium ion. This process has the potential to develop novel pathways for producing value-added chemicals through the efficient utilization of water-stable acyl fluorides and the unique 1,3-chelation concept in biomass.

Experimental Section

Synthesis of acyl fluorides 2: To a stirred solution of acyl chloride (0.5 mmol, 1.0 equiv.) in distilled acetone (2.5 mL, 0.2 M), potassium fluoride (1.0 mmol, 2.0 equiv.) was added. The reaction mixture was then stirred at room temperature for a predetermined period. The precipitated potassium fluoride was filtered through filter paper, and the reaction mixture was then subsequently purified according to the following procedure to yield acyl fluoride **2**. Purification steps: the residues **1a–c** were evaporated under reduced pressure to yield **2a–c** without further purification; the residues **1d–1p** were subjected to chromatography on silica gel with hexanes as the eluent, and subsequently evaporated under reduced pressure to obtain pure forms of **2d–p**; the residues **1q–r** were evaporated under reduced pressure,



Scheme 9. Gram-scale experiment (benzoyl fluoride **2a** was purified prior to use).

washed with deionized water, and then filtered through filter paper. The filtered solids were washed with acetone, and the resulting filtrates were evaporated under reduced pressure to yield pure forms of **2q–r**.

Synthesis of acylated product 4–6, and 8: To a stirred solution of glycerol (1.0 mmol, 1.0 equiv.) in acetone/H₂O (v/v = 9/1, 5 mL, 0.2 M), K₂CO₃ (1.2 mmol, 1.2 equiv.) and acyl fluoride **2** (2.0 mmol, 2.0 equiv.) were added, and the mixture was stirred at room temperature for 24 h. After completion of the reaction, the mixture was extracted with ethyl acetate. The extracted organic was dried over anhydrous magnesium sulfate, filtered, and concentrated *in vacuo*. The crude residue was purified by flash column chromatography on silica gel and eluted with ethyl acetate/hexane (1:3 v/v) to give the corresponding acylated products (**4–6**, and **8**).

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